



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year: 2022-2023(Odd Semester)

Degree, Semester& Branch: VII Semester B.E Computer Science and Engineering

Course Code & Title: CS8792 Cryptography and Network Security

Name of the Faculty member: Dr.R.Venkatesh, AP(SG)/CSE

Date: 22.09.2022

Innovative Practice Description

- **Unit / Topic:** III/Primality Testing
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO18
- **Activity Chosen:** Flipped Class Room
- **Justification:**

Flipped classroom is a pedagogical approach in which traditional content delivery moves to an individual self-learning process, and the group activity is transformed into a dynamic, interactive learning environment where the instructor just guides the learners to apply their creative thinking to provide a fruitful solution in the course content. It provides more benefits than traditional direct content delivery. With the advantage of students learnt this topic in Number theory

- **Time Allotted for the Activity:** 20mins
- **Details of the Implementation:**

(i) Materials for the Activity:

The materials for the preparation of the students were shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.

(ii) Formation of Groups:

As per the number of sub-topics, student groups were created as below:

- Number of sub-topics: 03
- Class Strength: 45
- Number of groups: 08
- Members per group: 06

(iii) Topic Allocation: For three sub-topics, students were divided into eight groups, and topics were allotted in advance. The students must prepare a 5-minute presentation, including the topic learned, 3 minutes for questions and answers, and a description of each member's participation.

(iv) Topic Delivery: The main objective of the flipped classroom activity is that each group will learn one concept, and the presentation helps the other members learn about the topic as well.

- **CO – PO / PSO mapping:**

CO	PO1
CO1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1
	3
Justification for correlation	Basic Computer Science Engineering and mathematics Knowledge for performing classical encryption is highly correlated

- **Images / Screenshot of the practice:**



Figure 1 – Students Discuss and Presentation their topics

- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

The students felt that it helped them to improve presentation skill and avoid stage fear. They utilized this for recapping things from number theory, which they have learnt in previous semester. They could get idea from learning sources which would help them to prepare the presentation

❖ ***Benefit of the Practice***

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of primality testing.

❖ ***Challenges faced in implementation:***

- ✓ Some of the students felt difficult to present because of lack of communication.
- ✓ Few groups done their presentation beyond their time limit.
- ✓ Few students in the group felt difficult to answer all the questions.

References:

- Flipped Learning Network (FLN). (2014) The Four Pillars of F-L-I-P™
- William Stallings, Cryptography and Network Security: Principles and Practice, PHI 3rd Edition, 2006.
- Mr.M.Gomathy Nayagam, “HDFS Concepts” in course “CS6703 Grid and Cloud Computing” dated:16.09.2019

Signature of Faculty Member

HOD-CSE